

ESCORT: Fine-Grained Urban Crime Risk Inference Leveraging Heterogeneous Open Data

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Abstract—Crime has been a complex social problem worldwide, impacting numerous individuals in both property and psychology, and affecting public safety as well. To prevent and avoid crime is of great importance for urban authorities and citizens. Crime prediction using various urban sensing data provides a promising paradigm to cope with this challenging problem. However, previous studies on crime prediction mainly focus on coarse-grained prediction at the region level, which fail in fine-grained road-level crime inference due to data sparsity problems. This article proposes a road-level framework for fine-grained crime risk inference leveraging heterogeneous open data. We first extract a set of features from relevant external datasets and historical crime records, i.e., temporal, spatial, and recurrence crime features. We then establish a spatio-temporal crime pattern using spatio-temporal features, and further infer road-level crime risk incorporating with recurrence crime features. The framework not only considers a useful spatio-temporal pattern but also considers a unique crime near-repeat phenomenon, which helps address the data sparsity. We further present an application *crime risk-aware route recommendation* to demonstrate the effectiveness of our framework. This application cannot be solved well with coarse-grained (e.g., region-level) crime risk results since it is inconvenient for people to travel while avoiding large regions with potential crime risks. Experiments with the real-world datasets from New York City show that our framework accurately infers road-level crime risk and outperforms other baseline methods.

Index Terms—Crime inference, crime risk, heterogeneous open data, route recommendation, urban computing.

I. INTRODUCTION

PUBLIC safety has been an increasingly critical and severe problem worldwide nowadays. As reported in [1], major crimes (e.g., violent crime and murder) kept increasing in cities worldwide during the past few years. Consequently, crime brings

tremendous damage to people's properties, psychological conditions, and lives. For example, crime in the USA during 2015 resulted in property losses up to \$14.3 billion [2]. Furthermore, crime impacts the sustainable development of societies [3]. Therefore, designing effective strategies for crime prevention and reduction is significant. From the perspective of local authorities, these strategies can help decrease possible crime with early warnings and promote public safety. From the perspective of citizens, these strategies can help avoid some possible crimes. For example, when a tourist arrives in an unfamiliar city and navigates to a place of interest, it is highly necessary for him/her to find a risk-aware route with a lower possibility of potential crime.

In the past decades, using heterogeneous open data to address pervasive computing applications has attracted a lot of attentions, e.g., mobile user sensing [4], [5], urban services [6], [7], and trading intelligence [8]. Recently, there have been many studies on crime prediction, using demographic data, socio-economic data, points of interest (POIs), etc. [9], [10]. These studies only considered static characteristics of crime events, which does not change over time, making it difficult to obtain good prediction due to the lack of consideration of crime dynamics. Some other studies exploited social network and mobility data in crime prediction at the community-level to reveal the dynamic characteristics of crime events [11], [12]. These studies can be used to understand the coarse-grained crime risk patterns but fail to generalize to road-level crime risk inference due to data sparsity problems, i.e., there are not enough samples in fine-grained levels. Thus, they cannot satisfy people's growing demand for fine-grained accurate crime prediction, such as at road-level. It is inconvenient for people to travel while avoiding such large regions with potential crime risks. It also leaves room for authorities to improve working efficiency. Thus, it is necessary to conduct crime risk prediction at a more fine-grained level. With accurately inferred road-level crime risk, local authorities can optimize police resource schedules toward specific roads. So the patrol efficiency can be improved, and public safety can be promoted. Citizens can make better preparation for potential criminal activities accordingly, e.g., adjusting their decisions to avoid highly risky routes.

Therefore, we attempt to conduct a fine-grained crime study to infer road-level crime risk, which is not easy. Criminal studies have indicated that the occurrence of crime events would be closely related to time and space at a coarse-grained level [13], [14]. We also conduct visualization analysis to further find their correlation in spatial and temporal dimensions at a fine-grained level. We observe that crime density difference varies with locations from Fig. 1(a) and that crime occurrences in one specific road vary with time changes from Fig. 1(b). Thus, it is

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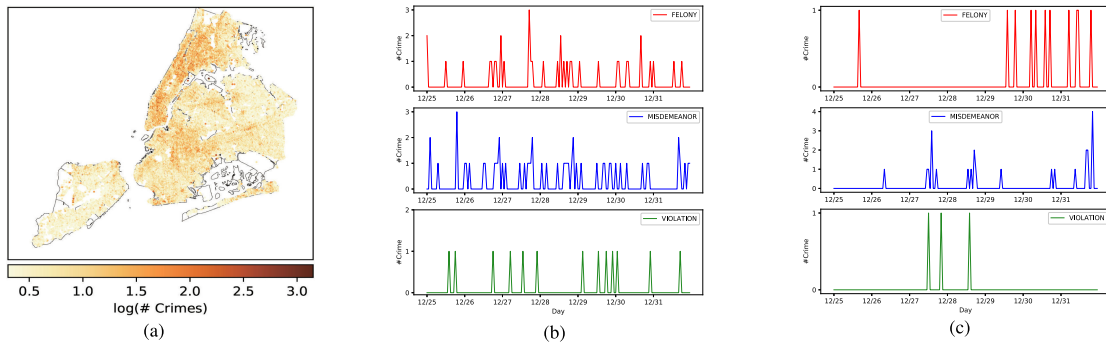


Fig. 1. Spatial and temporal distribution of crime in New York City during year 2015. (a) Spatial distribution. (b) Temporal distribution in Rawlins Ave, Bronx. (c) Temporal distribution in Schenck Ave, Brooklyn.

possible to infer fine-grained crime accurately by incorporating spatial and temporal crime-relevant factors. Fortunately, with the development of ubiquitous sensing technologies, we can obtain various large-scale urban open data [15]. With more smart environments [16] in the future, more ubiquitous urban data would be sensed and collected. These heterogeneous data can be exploited to find crime-relevant factors and understand crime risk from a more profound and broader perspective. For example, meteorological data has been investigated to be relevant to crime, e.g., weather and temperature information [17], [18]. Thus, big urban data provides us an opportunity for accurate fine-grained crime inference.

We propose to infer fine-grained crime risk at road-level leveraging large-scale urban data and then generate crime risk-aware route recommendation. First, we attempt to predict crime occurrences in roads. One intuitive method is to extract a set of relevant features, including spatial contextual features (e.g., POIs) and temporal contextual features (e.g., meteorological data). Then, prediction models are built with extracted features for road-level crime occurrences inference. However, such methods are unable to work well in fine-grained crime risk inference scenarios, e.g., they can hardly predict crime occurrences accurately in some roads due to the data sparsity. Fig. 1(b) and (c) illustrates an intuitive comprehension to such difficulty. We find that crime events of different severity levels in Rawlins Ave, Bronx, have obvious regularities. But it is difficult to quickly find regularities for crime events in Schenck Ave, Brooklyn, with very sparse values. Hence, it is not always effective to directly adopt a prediction model with spatio-temporal features to infer crime risk at the road-level.

Meanwhile, we find one type of sociological phenomena that can help improve the fine-grained crime inference accuracy, i.e., *near-repeat phenomenon*. This phenomenon refers to possible reoccurrence of crime events in close proximity in both time and space [19]. It can be a predictor for possible future crime occurrences, especially sometimes the spatio-temporal pattern is not effective for inference. For example, occurrences of crime events will be declined rapidly if more police officers are patrolling on the streets, which would only appear on some special and important occasions due to the constraint police resource. Here, using only spatio-temporal patterns cannot interpret the possible underlying reasons for the rapid short-term changes. In contrast, the near-repeat phenomena can help predict future criminal activities based on recent crime occurrences that happened in near space. Therefore, we can improve the fine-grained crime inference accuracy by incorporating recent occurrence of crime events with spatio-temporal pattern of crime events.

In this article, we propose a road-level framework for fine-grained crime risk inference leveraging heterogeneous urban open data. In particular, we first extract three categories of features from various external datasets and historical crime records, i.e., temporal features (e.g., date, time, and meteorological data), spatial features (e.g., POIs, taxi mobility, demographic data and police station distribution, and recurrence crime features). We then feed these spatio-temporal features into a cotraining model to achieve the spatio-temporal crime knowledge in roads. After that, we incorporate the spatio-temporal crime knowledge with recurrence crime features to infer the final knowledge of crime risk using a Bayesian model. Based on the crime risk of roads, we present an application as risk-aware route recommendation to find a set of Pareto-optimal routes with minimum overall distance and crime risk. The main contributions of our article include the following.

- 1) To the best of our knowledge, this is the first work on crime risk inference at road-level leveraging heterogeneous open data and provide crime risk-aware route recommendations. We can accurately infer fine-grained crime risk by fusing large-scale urban sensing data and generate optimal risk-aware routes based on inferred road-level crime risk.
- 2) We propose a two-phase framework to infer fine-grained crime risk. Using various urban data, we first extract a set of features, including temporal, spatial, and recurrence crime features. A cotraining and Bayes (CoBayes)-based method is proposed to infer the crime risk in roads. CoBayes estimates the spatio-temporal crime occurrences by the cotraining model using spatio-temporal features and infers the final knowledge of crime risk incorporating the recurrence crime features and the spatio-temporal crime estimation by using a Bayesian model.
- 3) We present a useful application, i.e., risk-aware route recommendation, which provides a set of risk-aware optimal routes with minimum overall distance and crime risk.
- 4) We evaluate our framework with real-world datasets. Results show that our framework can accurately infer fine-grained crime risk in roads and effectively generate risk-aware optimal routes, outperforming other compared methods.

II. RELATED WORK

In this section, we examine the work on crime prediction at first, and then investigate existing literature on route planning.

A. Crime Prediction

Recently, there have been many studies exploiting various influential factors for crime prediction at different levels. Examples include demographic data (e.g., education, income, ethnic diversity, etc.) [9], [10], economic data (e.g., real estate) [20], mobility data (e.g., location-based social network (LBSN), taxi trip, mobile network) [11], [12], [21], [22], and environment data [11], [12], [23]. These work can benefit us to conduct studies for accurate crime prediction with the help of multisource data. From the perspective of geographical scales, the existing work can be mainly categorized into two levels, i.e., community-level and grid-level. There have been several previous studies on the crime prediction at the community-level. Wang *et al.* [11] employed two models to infer the yearly crime rate of different communities of Chicago, with the utilization of POI and taxi flow data. Fabusuyi [20] conducted studies to analyze the relationship between crime and real estate through a case study of East Liberty. These previous studies mainly conducted crime prediction on a coarse-grained level. There also have been some crime prediction studies on a fine-grained level, i.e., grid-level or hotspot-level. Here, *grid* or *hotspot* usually refers to nonunique-defined areas of high crime concentrations. Yang *et al.* [12] proposed a new platform CrimeTelescope to forecast crime hotspots with the utilization of urban and LBSN data and visualize the hotspots on an interactive interface. Wang *et al.* [22] proposed a Twitter-based criminal incident prediction method, which employed an automatic semantic analysis to understand Twitter posts. Gerber [21] adopted GPS-tagged tweets and kernel density estimation (KDE) to predict the major 25 crime types and found that using tweets data can be helpful to improve the accuracy among 19 of 25 crime types. However, these studies cannot satisfy people's demand for a more fine-grained crime prediction, e.g., road-level. With the road-level crime prediction results, citizens can use this information to make risk-aware traveling decisions, and authorities can optimize working schedules to reduce and prevent crime effectively. In this article, we attempt to address the problem of fine-grained crime risk prediction leveraging heterogeneous data.

B. Route Planning

The route planning problem has been studied by a number of previous literature [24], [25], including routing for vehicles and for tourists. Sarker *et al.* [26] developed a multiobjective routing system, aiming to minimize energy consumption, traveling time, charging cost, and range anxiety for vehicles. These previous studies are mainly for drivers to find suitable routes with defined goals. There also have been a lot of studies on route recommendations for people, e.g., tourists. Chen *et al.* [27] developed a new framework to recommend routes for tourists and interested peoples, with the utilization of geo-tagged images and check-in data from LBSN. A heuristic algorithm was designated with the purpose of maximum scenic view score under user-specific constraints. These works conducted studies to recommend routes for tourists with designated objectives. Closer to our article, Galbrun *et al.* [28] detected the crime hotspots using crime data from Chicago and Philadelphia, then predicted the crime risk of each road, and finally provided risk-aware route recommendations. In this article, we leverage multisource heterogeneous open data for crime risk computation at the road-level and then provide optimal routes recommendation for all users with the purpose of minimum distance and crime risk.

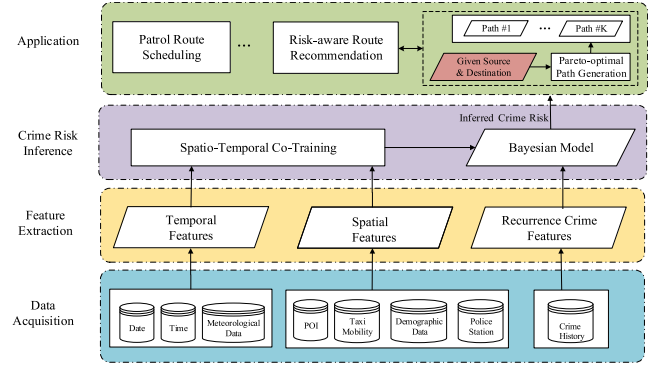


Fig. 2. Overview of ESCORT framework.

III. PRELIMINARY AND FRAMEWORK OVERVIEW

Definition 3.1 (Road network): A road network is defined as a geographical region consisting of a finite set of nodes $\mathcal{N} = \{n_1, n_2, \dots, n_z\}$ and a finite set of edges $\mathcal{E}$ where graph edge e_{ij} denotes a road segment connecting n_i and n_j . For simplification, we use *road* to represent the *road segment*.

Definition 3.2 (Crime event): A crime event that happened in coordinate $(lat, long)$ of road r at time t with severity level s in crime type c is denoted by 5-tuples as $(c, r, t, lat, long, s)$.

Definition 3.3 (Crime risk): Inspired by the work [29], we think that crime risk can be reflected by the severity levels of crime events occurred. Thus, we define crime risk $\Phi(r, t)$ as the summation of severity levels of all crime events occurred in road r at time t .

We propose ESCORT, a two-phase framework to infer fine-grained crime risk at road-level. Fig. 2 illustrates an overview of the proposed framework ESCORT with several applications. In this article, we choose a risk-aware route recommendation to present the effectiveness of our framework. Note that the inferred road-level crime risk as well can be used for other applications, e.g., patrol route scheduling. We first collect a large amount of relevant heterogeneous data, e.g., crime data, demographic data, taxi mobility data, POIs, and meteorological data. We then extract a set of features, including temporal, spatial, and recurrence crime features. In the crime risk inference phase, we first feed temporal and spatial features into a cotraining model to calculate spatio-temporal estimation of crime occurrences in roads. Incorporating this spatio-temporal crime knowledge and recurrence crime features jointly, we infer fine-grained crime risk using a Bayesian model. With these inferred crime risks on roads, given the source location and the destination location, we can provide a set of Pareto-optimal crime risk-aware routes recommendation suggestions considering both route distance and potential crime risk jointly. We elaborate on details of the framework in the following sections.

IV. FINE-GRAINED CRIME RISK INFERENCE

In this section, we present our ESCORT framework for crime risk inference. We first extract relevant features, including temporal, spatial, and recurrence crime features, and then we present the two-step crime risk inference method.

A. Feature Extraction

1) *Temporal Features:* Previous studies have shown that temporal contextual factors related to crime usually contain: *date*

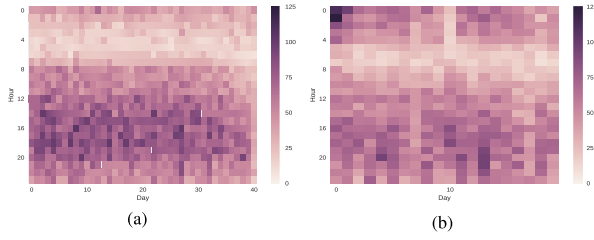


Fig. 3. Crime occurrences in New York City during two months (November 1, 2015 to December 31, 2015). (a) Weekdays. (b) Weekends and holidays.

TABLE I
DURATION GROUPS FOR MODELING TEMPORAL CONTEXT

Day type	Duration Groups	Time span	Abbr.
Weekdays	Wee hours	1:00 - 7:00	w1
	Day hours	7:00 - 12:00	w2
	Active hours	12:00 - 21:00	w3
	Night hours	21:00 - 1:00	w4
Weekends/Holidays	Day hours	5:00 - 10:00	h1
	Night hours	10:00 - 5:00	h2

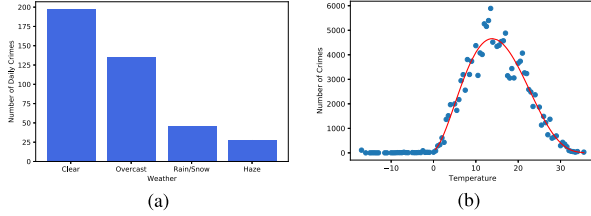


Fig. 4. Number of daily crimes in meteorological contexts. (a) Weather condition. (b) Air temperature.

and time, weather condition, and air temperature. Using crime data from New York City Police Department (NYPD) [30] and meteorological data from the Weather Underground [31], we analyze the impact of temporal contextual factors as follows.

a) Date and time: Intuitively, the occurrence of crime would be time-variant. For example, there are very few people outside during the wee hours, leading to a very low possibility of crime commitment. We present the daily crime occurrences of a period sample (November 1, 2015–December 31, 2015) in Fig. 3. We observe that the crime occurrences differ between weekdays and weekends/holidays and vary with different temporal durations as well. Hence, we divide the 24 hours of different day types into different temporal durations, as shown in Table I. Here, the date and time feature f_t of road r is one-hot representation with six temporal durations.

b) Weather condition: As indicated in previous studies [17], crime occurrences may be different with various weather conditions. For example, the close-range visibility would be obscure during *haze* days, which would hinder potential crime commitments. During *clear* days, many people would choose to go out, which brings a higher possibility of some crime occurrences, e.g., petit larceny. Thus, we utilize the weather condition data during the year of 2015 and present the average daily crime occurrences under four categorized weather conditions in Fig. 4(a). We observe that the average daily crime occurrences vary in weather conditions, indicating that the weather condition should be treated as a temporal feature for crime prediction. We represent the weather condition features f_w of road r as a one-hot representation.

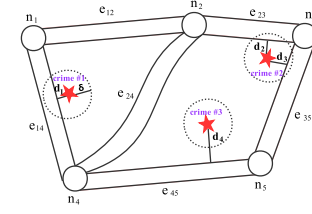


Fig. 5. Example of road network.

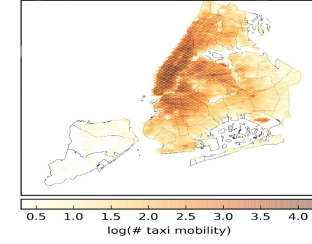


Fig. 6. Taxi mobility distribution.

c) Air temperature: We then study the relationship between air temperature and crime occurrences since the air temperature may have a correlation with crime occurrences [17], [23]. In cold and hot days, more people would prefer staying indoors, which leads to a lower possibility of crimes. In warm days with many people outdoors, some crimes would have a higher possibility to occur, e.g., illegal parking and larceny. Thus, we use air temperature data during the year of 2015 and show the relationship in Fig. 4(b). We observe an obvious relationship with a curve fitting between the number of crimes and air temperature values. Since air temperature is real-valued, the air temperature feature f_a is represented as a real number after normalization.

2) Spatial Features: Note that we focus on crime risk inference in roads. If a crime occurs far away from any road, we would dismiss it, e.g., *crime #3* in Fig. 5. If the distance between a crime and the nearest road is less than a threshold δ , e.g., *crime #1*, we believe this crime would also pose threats to people. If a crime may conduct potential threats to several roads, e.g., *crime #2*, we assume that this crime would induce influence on these roads. Intuitively, since different environments may have different socio-economic statuses, functionalities, and geographical structures, the occurred crime may vary significantly [9], [10]. We analyze the spatial features as following.

a) Taxi mobility: As indicated in previous work [11], occurrences of crime can be affected by human mobility. For example, more crowd of people in the public may increase the possibility of crime commitment, e.g., larceny. Here, we adopt taxi trip data as the representative of human mobility due to limited data access. We extract the pickup and dropoff data with the number of passengers of each trip and present the taxi mobility in a heatmap, as shown in Fig. 6. From this figure and Fig. 1(a), we observe that taxi mobility distribution hotspots have many overlaps with crime distribution hotspots. Hence, we extract the taxi mobility feature $f_m = (f_m.\text{pick}, f_m.\text{drop})$. Specifically, we define $\mathcal{A}$ as the set of pickup trips, $\mathcal{B}$ as the set of dropoff trips, and g_i as number of passengers of trip i , $i \in \mathcal{A} \cup \mathcal{B}$. We then compute these features shown as following:

$$f_m.\text{pick} = \sum_{i \in \mathcal{A}} |g_i| \quad (1)$$

$$f_{m,\text{drop}} = \sum_{i \in B} |\{g_i\}|. \quad (2)$$

b) POI: For a road r , the distribution of POIs in this road may represent this road's functionality. For instance, more entertainment facilities like night clubs in one road may attract more drunken brawls. We calculate the distribution of POIs across ten categories [32]. We define $\mathcal{O}_i$ to represent the i th category of POIs and compute the *term frequency-inverse document frequency* (TF-IDF) [33], [34] to represent the significance of each POI. Specifically, for r , its POI features $f_P = (tf_1, tf_2, \dots, tf_{10})$, where tf_i denotes the TF-IDF value of $\mathcal{O}_i$ in r . The value of tf_i can be computed as follows:

$$tf_i = \frac{n_i}{N} \times \log \frac{|\{r\}|}{|\{r | \mathcal{O}_i \in r\}|}. \quad (3)$$

c) Demographic characteristics: As well acknowledged, crime occurrences in one region would be affected by the social-economic environments [10]. Due to the protection of personal privacy, we obtain coarse-grained demographic data (i.e., community-level) rather than the detailed data at road-level as we focus on. Thus, we divide all roads into two subsets, one subset $\mathcal{S}$ consisting of roads with two terminals in one same community and the other subset $\mathcal{D}$ consisting of roads with two terminals in two different communities. More specifically, for $r \in \mathcal{S}$, there exists one specific community $\mathcal{C}_r$ with two terminals $r.n1$ and $r.n2$ both in $\mathcal{C}_r$. Thus, in $\mathcal{S}$, all roads share the same demographic characteristics. For r in $\mathcal{D}$, there exist two communities $\mathcal{C}_{r,1}$ and $\mathcal{C}_{r,2}$ which $r.n1$ and $r.n2$ lie in, respectively. We define the demographic characteristics of r as a weighted sum of demographic characteristics of $\mathcal{C}_{r,1}$ and $\mathcal{C}_{r,2}$. Demographic characteristics of r , which are defined as $\mathcal{D}_r$, can be calculated as (4). w_j equals the ratio of the length of the road segment part located in $\mathcal{C}_{r,j}$ to the length of r , $j = 1, 2$. Here, demographic characteristics include average income, education level distribution, average age, and population density. We define education level distribution as a percentage of different education levels, such as the percentage of people with a bachelor's degree. Thus, the demographic characteristics features $f_d = (f_d.\text{inc}, f_d.\text{edu}, f_d.\text{age}, f_d.\text{dens})$. All demographic characteristics data should be normalized

$$\mathcal{D}_r = \begin{cases} \mathcal{D}(\mathcal{C}_r) & r \in \mathcal{S}, r.n1 \in \mathcal{C}_r, r.n2 \in \mathcal{C}_r \\ \sum_{j=1}^2 w_j \times \mathcal{D}(\mathcal{C}_{r,j}) & r \in \mathcal{D}, r.n1 \in \mathcal{C}_{r,1}, r.n2 \in \mathcal{C}_{r,2}. \end{cases} \quad (4)$$

d) Police stations: Intuitively, occurrences of crime would be affected by the police station distribution. For instance, there may be less crime that happened near the police stations or regular patrol routes. Therefore, we take the police station's distribution into account. We define q to represent a police station, $\mathcal{Q}$ as the set of q s, and $\text{dist}(i, j)$ to represent the distance between i and j . For r , we compute the distance between r and the nearest police station as $\min_{q \in \mathcal{Q}} \{\text{dist}(q, r)\}$ and normalize it to represent the police station influence feature f_s .

3) Recurrence Crime Features: Crime occurrences would also be affected by the nearby historical crimes, i.e., the near-repeat phenomenon. This phenomenon indicates that crime events may reoccur in close proximity in both time and space [19]. From the perspective of this theory, crime occurrences of one region are not only influenced by historical local crime occurrences but also affected by the historical crime occurrences of nearby regions. Thus, the occurred crime events would become a predictor for future criminal activities.

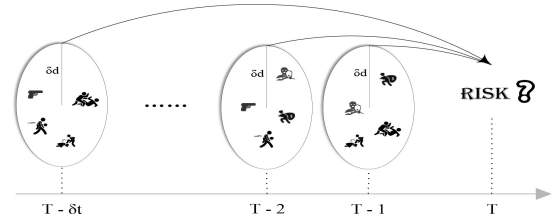


Fig. 7. Recurrence influence.

We define *recurrence crime feature* to represent this near-repeat influential factors for future crimes. We use the recent crime records that occurred in near space and time to estimate the near-repeat influence, as shown in Fig. 7. We hope to infer the recurrence crime risk from historical crime records during close proximity in space and time. Let δ_d denote the threshold of near space and δ_t denote the threshold of near time gap. We first define $v(c, r, t)$ as the occurrences of recurrence events of crime c in road r at duration t , and then define nearby recurrence crime distribution $D(c, r, t) = \{v(c, r, t') | r' \in E(r, \delta_d), t' \in [t - \delta_t, t]\}$ (D for short), as all $v(c, r, t)$ in the near δ_d space $E(r, \delta_d)$ during the past δ_t time. This can be treated as the recurrence crime feature that considers nearby crimes. The detailed process of this kind of feature is elaborated in Section IV-B2.

B. Crime Risk Inference

We aim to infer the road-level crime risk using large-scale heterogeneous urban data. The crime occurrences of type c at time duration t in road r is defined as $k(c, r, t)$ (k for short). We first adopt a cotraining style paradigm to calculate spatio-temporal estimation of k , using extracted temporal and spatial features. We then feed the spatio-temporal estimation and recurrence crime occurrences into a Bayesian model so as to update a posterior knowledge of k and infer the final crime risk in roads.

1) Spatial-Temporal Cotraining: We try to compute spatio-temporal estimation with a cotraining paradigm. Cotraining is a typical multiview learning approach employing two independent models for inference. Since the spatial and temporal contextual features are extracted from heterogeneous datasets, which cannot be considered equally [7], we choose the cotraining paradigm for this problem. However, in particular, cotraining is widely adopted in classification tasks while less addressed in regression problems. To effectively employ cotraining strategies in our problem which is a regression problem, we propose a semisupervised learning method based on cotraining regression (CoReg) [35].

We first fuse the temporal and spatial features by a concatenation operation as $F = (f_t, f_w, f_a, f_m, f_p, f_d, f_s)$. These features have been normalized to keep the same scale of all features. In the spatio-temporal CoReg paradigm, we input the concatenated spatio-temporal features F into two regressors, respectively, and iteratively add the instances which can bring less mean squared errors (MSEs) to the training dataset. Specifically, there are two datasets, i.e., labeled set L and unlabeled set U . $L = \{(x_1, y_1), \dots, (x_{|L|}, y_{|L|})\}$. Our objective is to estimate the real-valued output of each sample in U . We first initialize two sets L_1 and L_2 , both of which are the same as L . We also use two regressors for the cotraining paradigm. With randomly picked samples U' from U , we compute the estimated $\hat{y}$ of each x_u in U'

by the two regressors, respectively. After that, we employ three metrics here, i.e., MSE, mean absolute error (MAE), and root mean squared error (RMSE). We then compute these three errors of regressors with x_u and $\hat{y}$ as MSE'_u , MAE'_u , and $RMSE'_u$, respectively. We also compute these errors of regressors on labeled sets as $\delta s_u = MSE'_u - MSE_u$, $\delta a_u = MAE'_u - MAE_u$, and $\delta r_u = RMSE'_u - RMSE_u$, respectively.

We adopt δm as a weighted sum of three errors for the final selection. After all the computation of x_u in U' , the sample with the biggest positive δm should be treated as the most confident sample and then added to the labeled set L and deleted from U . Thus, spatio-temporal estimation $k(c, r, t)$ of crime occurrences of type c at time t in road r can be calculated.

2) *Bayesian Model*: As aforementioned, past crime events would cause possible recurrence of crimes in a short time. Based on the spatio-temporal crime estimation $k(c, r, t)$, we first update the estimation of $k(c, r, t)$ considering the possible influence of recurrence crimes and then compute weighted results of c with weight $w(c)$ as the risk level of c to compute the final crime risk $\Phi(r, t)$ of road r at time t (Φ for short).

According to the Bayesian theory, we formulate the crime occurrences as in (5). Specifically, we treat the $p(k)$ as the prior distribution for the crime occurrences in road r at time duration t . With the recently happened crime events D , we aim to estimate the posterior probability of crime occurrences in road r at time duration t

$$p(k|D) = \frac{p(D|k) \times p(k)}{p(D)}. \quad (5)$$

According to the aforementioned cotraining paradigm, we obtain the k data. Hence, we utilize an estimation for $p(k)$, as shown in (6). $\hat{p}(k)$ becomes larger with higher k in t

$$\hat{p}(k) = \frac{k}{\sum_{t \in T} k}. \quad (6)$$

As studied in [36], we set δ_d as 400 m and δ_t as 1 week. Due to the low-frequency characteristics of crimes, we assume that the impacts of these recurrence crime events are independent for simplification. Thus, we need to compute the $p(v|k)$ before estimating $p(D|k)$. The smoothing maximum likelihood estimate of the probability $p(v|k)$ is as follows:

$$\hat{p}_{ml}(v|k) = \begin{cases} \frac{\sum_{t \in T} v}{\sum_{r \in E} \sum_{t \in T} v}, & v > 0 \\ \frac{\sum_{r \in E} \sum_{t \in T} v}{\sum_{r \in E} \sum_{t \in T} v}, & \text{otherwise} \end{cases} \quad (7)$$

$$\hat{p}_{ml}(D|k) = \prod_{v \in D} \hat{p}_{ml}(v|k). \quad (8)$$

The distribution of c in road r during δ_t can be treated as a Poisson process, as shown in (9). Here, $\lambda(c, r, t)$ is defined as a random variable reflecting the mean occurrences of c in terms of road r and time duration t . The value of $\lambda(c, r, t)$ varies with different crime type c , road r , and time duration t

$$\hat{p}(D) = \prod_{v \in D} \hat{p}(v) \sim \prod_{v \in D} \text{Poisson}(\lambda_v). \quad (9)$$

Altogether, we have the final expression. The inferred crime occurrences of crime c in road r at time duration t would be k with the largest probability

$$\hat{p}_{ml}(k|D) = \frac{\hat{p}_{ml}(D|k) \times \hat{p}(k)}{\hat{p}(D)} \quad (10)$$

$$\hat{k}_{ml} = \arg \max_k p(k|D). \quad (11)$$

After the crime occurrences inference, we can compute the final risk in road r at time duration t with the utilization of weights assigned to each crime type, shown as following:

$$\Phi(r, t) = \sum_c w(c) \times \hat{k}_{ml}(c, r, t). \quad (12)$$

V. APPLICATION TO RISK-AWARE ROUTE RECOMMENDATION

In this section, we will conduct a crime risk-aware route recommendation based on the inferred crime risk of roads. Solving this task needs crime risk information for each road. Thus, our fine-grained crime risk inference can provide useful crime risk information of different time durations in each road to help solve this task. In this route recommendation, the objective is to find optimized routes for passengers, given a source location (S) and a destination location (D). Specifically, we formulate the risk-aware routing problem as a biobjective optimization problem to minimize the route length and risk. We then present the solution of this biobjective optimization problem by finding a Pareto front set.

We take Fig. 5 as an example. Given a source location node n_1 and a destination location node n_5 , there are several different choices from n_1 and n_5 , such as $n_1 \rightarrow n_2 \rightarrow n_3 \rightarrow n_5$, or $n_1 \rightarrow n_4 \rightarrow n_5$. These choices refer to a set of routes $\mathcal{P}$. We define $\tau_{i,j}^k$ as a decision variable for edge $e_{i,j}$ and route $p^k \in \mathcal{P}$ to indicate whether $e_{i,j}$ is involved in route p^k , as shown as the following:

$$\tau_{i,j}^k = \begin{cases} 1, & e_{i,j} \in p^k \\ 0, & e_{i,j} \notin p^k \end{cases} \quad (13)$$

We aim to minimize the crime risk of involved roads and simultaneously minimize the overall length of these roads. Thus, we formulate the biobjective route optimization problem as an integer-programming problem. First, we define the length of a specific route by summing up the distance of consecutive edges of the route. Let $\Xi(i, j)$ denote the distance of edge $e_{i,j}$. $\Xi(n_i, n_j)$ can be calculated by the coordinates of the two nodes n_i, n_j and edge $e_{i,j}$. Second, we quantify the crime risk of a specific route by summing up the inferred crime risk of consecutive edges of the route. We define $\Phi'(n_i, n_j, t)$ as the risk of edge $e_{i,j}$ at time t . Here, the definition of Φ' is different from Φ , as it represents the crime risk measure between any two nodes rather than two adjacent nodes in one road. We also consider that people would have an upper threshold to endure the crime risk when having options to select the routes. Hence, we define $\delta_{\Phi'}$ as the upper threshold

$$\Xi_k(S, D) = \sum \tau_{i,j}^k * \Xi(n_i, n_j) \quad (14)$$

$$\Phi'_k(S, D, t) = \sum \tau_{i,j}^k * \Phi'(n_i, n_j, t). \quad (15)$$

Finally, we formulate the following biobjective optimization problem by applying the two aforementioned objective functions. Here, these two objective functions may not have a positive correlation. The route with the shortest distance between S and D may have higher crime risk in some occasions, indicating that there may be a tradeoff between the distance and the crime risk. We would present the solution for this biobjective optimization problem as follows:

$$\min_{p^k \in \mathcal{P}} (\Xi_k(S, D), \Phi'_k(S, D, t)) \quad (16)$$

subject to

$$\tau_{i,j}^k = \{0, 1\} \forall i, j, k \quad (17)$$

$$0 \leq \Phi'_k(n_i, n_j, t) \leq \delta_{\Phi'} \forall n_i, n_j, t. \quad (18)$$

To solve the aforementioned optimization problem, we use the ϵ -constraint method to generate Pareto front for this biobjective problem. The Pareto-optimal solutions are feasible here since they pay more attention to the tradeoffs between these two objectives.

Background: Before introducing the concept of Pareto-optimal, we first briefly introduce the concept of Pareto dominance. Given two routes p^1 and p^2 , we define p^1 dominates p^2 if and only if there has been at least one objective that p^1 has lower value than p^2 ; meanwhile, p^2 do not achieve lower value than p^1 in both of the two objectives. A route p can be called Pareto-optimal if there does not exist any other route dominating p . Then, Pareto set denotes the set of all Pareto-optimal variable vectors, and Pareto front denotes the set of all Pareto-optimal objectives.

To generate Pareto front of this biobjective problem, we employ an adaptive ϵ -constraint method [26], [37]. This method treats the biobjective optimization problem as a searching problem, meanwhile considering each objective function separately. Specifically, we first solve the two objective functions separately and find the corresponding shortest routes, respectively. These consequent values would be defined as their lower limits (ϵ'). Then, we determine the coordinates for the searching grid by the function values and store in the matrix e . We also define P as the set of existing Pareto-optimal points and Q as the searching space. In each iteration (steps 1–17), we obtain a new Pareto-optimal point x with the utilization of $\text{opt}(f, \epsilon, \epsilon')$ and check whether x dominates P . If so, we add x to P , the set of existing Pareto-optimal points. Meanwhile, Q would be updated by adding $[f(x), \epsilon]$, and the grid would be updated by $[\epsilon, \epsilon']$ consequently.

VI. EXPERIMENTAL EVALUATION

In this section, we first introduce the used datasets, implementation and metrics, as well as compared approaches. Then we evaluate the performance on crime risk inference and risk-aware route recommendation. Besides, we also conduct case studies to demonstrate the effectiveness of our methods.

A. Datasets

We evaluate our methods using urban data of New York City and summarize the data in Table II.

1) *Crime*: We retrieve crime data from historical complaint data provided by NYPD [30], which record all valid crime events that residents reported to NYPD. The dataset contains all valid crime types, such as larceny, assault, burglary, etc.

2) *Meteorological Data*: We collect meteorological data of 2015, consisting of weather conditions, temperature, humidity, barometer pressure, and wind strength. The dataset is crawled from a public website [31] sampling every hour. We extracted the weather condition and temperature to represent the influence of the meteorological aspect.

3) *Road Network*: We collect the road network data from OpenStreetMap [38]. The road network is composed of 5 531 122 intersections and 586 200 roads. These roads include highways, motorways, tunnels, bridges, and so on.

TABLE II
DETAILS OF DATASETS

Data Sources	Properties	Values
Crime	# categories	312
	# instances	148834
Meteorological data	# attributes	9
	# instances	10747
Road network	# intersections	5,531,122
	# streets	586,200
POIs	# categories	10
	# instances	69,714
Taxi mobility	# green taxi trips	19,233,765
	# yellow taxi trips	77,080,575
Demographical data	# communities	59
	# instances	195,742
Police stations	# boroughs	5
	# precincts	77
Data collection period		08/01/2015 00:00 - 08/31/2015 23:59
City		New York City
Geographic coverage area	Southwest:	[40.494852, -74.257124]
	Northeast:	[40.912324, -73.714674]

4) *POIs*: We collect 69714 POI data of New York City from [32]. These POIs belong to ten first-tier categories respectively. We extract the POI distribution of each road in this article to represent the spatial importance of POIs in roads.

5) *Taxi Mobility*: We obtain a taxi mobility dataset from NYC Taxi and Limousine Commission [39]. The dataset contains all the trip records of yellow taxis and green taxis in New York City. We only extracted pickup and dropoff information with locations and reported passenger counts to represent the taxi mobility characteristics.

6) *Demographical Data*: We collect community district-based demographical profiles from the American Community Survey 2015 by Census Bureau of New York City [40].

7) *Police Stations*: We collect the police station distribution data from the New York Police Department [41], with 77 precincts in 5 boroughs. We extracted the location information of precincts to measure the distance between locations where crime events occurred and their nearest precincts, which can represent the influence by police stations.

We here demonstrate the crime risk situation in different time spans of New York City using data of August 18, 2015, as shown in Fig. 8. The red dotted circles in each figure present the crime risk difference between this duration and other durations. From the figure, we can find that crime risk varies with different time spans, which is consistent with our aforementioned analysis. Here, we do not present the crime risk situation during wee hours on weekdays and day hours on weekends/holidays since there has been sporadic crime records citywide. As aforementioned, potential motivated criminals are often attracted to some vulnerable geographical locations by desired objectives (e.g., money and property) and insufficient security resources (e.g., absence of guardians or unpatrolled roads) [42], [43]. For targeted locations, they would experience different crime risks during different time periods. For example, a dense community may have many cars parked roadside. These cars in the streets may become targets for potential criminals for larceny. As shown in Fig. 8(a), during the period 10:00–5:00 on weekends, residents are more likely to go out to shopping malls, museums, or parks. Thus, more vehicles would be parked in the streets, which leads to a higher crime risk during this certain period. During other periods, e.g., 12:00–21:00 on weekdays, neighborhoods with many dual-income families would also be potential targets for criminals for burglary since there would be few people at home.

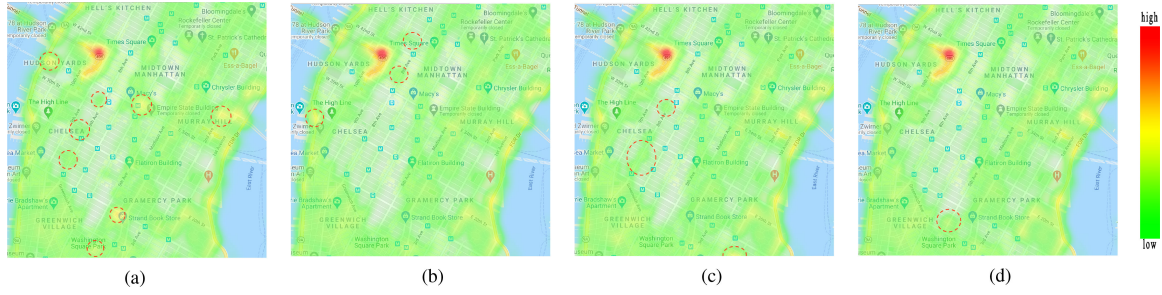


Fig. 8. Overall crime risk situation in New York City. (a) 10:00–5:00 on weekends. (b) 7:00–12:00 on weekdays. (c) 12:00–21:00 on weekdays. (d) 21:00–1:00 on weekdays.

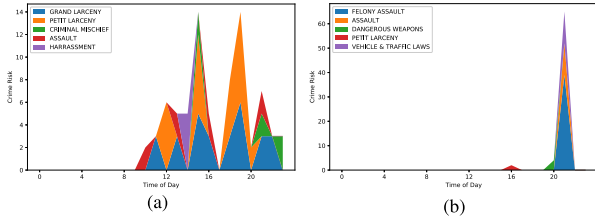


Fig. 9. Top five crime types changing over time. (a) Broadway (W 133rd St–W 135th St). (b) Baisley Boulevard.

We also compare the crime risk of the top five crime types varying over time at different roads, e.g., road Broadway (W 133rd St–W 135th St) and Baisley Boulevard. As shown in Fig. 9, we notice obvious differences. At road Broadway, there are frequent crime events during a day, with two spikes at daytime and night. Larceny is the most frequent crime event, indicating that there are more valuable properties in road Broadway. In contrast, at Baisley Boulevard, the most frequent crime types here are different from those at road Broadway, with more assault events. There are relatively many schools nearby and possibly more brawls happening among the teens from the schools.

B. Implementation and Evaluation Metrics

We evaluated our proposed framework with two steps, crime risk inference and risk-aware route recommendation. First, we extracted temporal and spatial features and fed them into a cotraining paradigm to infer the spatio-temporal knowledge of road-level crime risk. Moreover, we took into account the recurrence risk with location and time duration information and then updated the corresponding knowledge of crime risk on roads. We compared our proposed models with state-of-the-art approaches to verify the effectiveness of our models, with data from the last ten days as the testing dataset (i.e., hold-out samples) and data before that as the training dataset. In particular, after we divide the dataset into training datasets and testing datasets, we partition 10% of each training dataset as labeled dataset and the remaining 90% as an unlabeled dataset in CoReg process. One hundred runs of experiments have been conducted on each dataset, and the maximum numbers of iterations and pool size are set to 100 and 1000, respectively.

Second, after the crime risk inference, we can infer the crime risk of all roads, which helps us provide many applications. We chose the risk-aware route recommendation to demonstrate the applicability of the aforementioned study. We extracted the road network of New York City as a graph using the NetworkX

package in python. And then, we calculated the distance of edges using coordinates information and crime risk of edges. Given the source and destination locations, we provided several Pareto-optimal and nondominated routes with minimum distance and risk. Besides, we compared our method with the distance-optimal method and risk-optimal method to evaluate the advantages of our method.

We compared our crime risk inference results with ground truth dataset to verify the effectiveness of our proposed models. Specifically, if the crime risk of one road in a time duration is y_i with inference result $\hat{y}_i$, we quantitatively evaluate the performance of our models by employing the following metrics: MAE, mean relative error (MRE), and RMSE. Here, the ground truth value of the crime risk of road r during time t is calculated as the summation of severity levels of all crime events occurred in road r during time t . Specifically, smaller value of these metrics indicates better inference performance

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (19)$$

$$\text{MRE} = \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i} \quad (20)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N |y_i - \hat{y}_i|^2}{N}}. \quad (21)$$

C. Compared Approaches

We compared our proposed methods with several state-of-the-art approaches. For crime risk inference, we compared our models with the following approaches.

- 1) Least absolute shrinkage and selection operator (LASSO) [44]: It is a linear regression model with L1-norm regularizer, which can enhance the inference accuracy.
- 2) Ridge [45]: It is a linear regression model with L2-norm regularizer, which can also improve the effectiveness and efficiency of inference.
- 3) Support vector regression (SVR) [46]: This approach is a regression approach using the same principles as those in support vector machine (SVM) with minor differences.
- 4) Decision tree (DT) [47]: This regression approach is able to train a CART tree for the crime risk inference.
- 5) Random forest (RF) [48]: This approach is able to train an RF regressor to infer the crime risk.
- 6) Negative binomial (NB) [11], [49]: This approach is a popular statistical method for crime risk regression.

TABLE III
COMPARISON WITH APPROACHES W.R.T MAE, MRE, AND RMSE

Methods \ Metrics	MAE	MRE	RMSE
LASSO	1.113	0.814	1.686
Ridge	0.923	0.691	1.477
NB	0.907	0.759	1.423
LSTM	0.882	0.717	1.466
SVR	0.878	0.631	1.457
DT	0.852	0.609	1.475
RF	0.786	0.581	1.412
CoBayes	0.767	0.573	1.348

The bold values emphasize the advantageous of CoBayes method over other methods.

- 7) Long short-term memory (LSTM) [50]: This approach trains a three-layer neural network with LSTM units to infer future crime risk.
- 8) TBayes: This approach only exploits the temporal features to predict the crime risk and then update the crime risk combined with recurrence risk by using the Bayesian model.
- 9) SBayes: Similarly, this approach exploits the spatial features to infer the crime risk with recurrence influence through the utilization of the Bayesian model.
- 10) CoReg [35]: This approach employs the same co-training method of the proposed method *CoBayes* using the same spatio-temporal features for regression without consideration of recurrence risk and assign the same weights as *CoBayes* to infer the final crime risk of given road r at time duration t . The computation is as (22). The final crime risk $\Psi(r, t)$ is a weighted sum of spatio-temporal crime estimation $k(c, r, t)$

$$\Psi(r, t) = \sum_c w(c) \times k(c, r, t). \quad (22)$$

For risk-aware route recommendation, we formulated it as a biobjective optimization problem to minimize distance and risk and then provided nondominated routes. We compared our adaptive ϵ -constraint method with following approaches.

- 1) Distance optimal route (DOR): This approach can provide routes generated with minimum distance, with less consideration of crime risk of routes.
- 2) Risk optimal route (ROR): This method generates routes with minimum risk, with less attention on the distances.

D. Evaluation on Crime Risk Inference

We compare the performance of our framework with other methods in Table III. We observe that CoBayes achieves the best performance in terms of all metrics (MAE = 0.767; MRE = 0.573; RMSE = 1.348), outperforming other methods, especially in MAE. Concerning MAE, CoBayes obtains a substantial improvement, with around 31.09% higher than LASSO, 16.90% than Ridge, 15.44% than NB, 13.04% than LSTM, 12.64% than SVR, 9.98% than DT, and 2.42% than RF. This result verifies the effectiveness and efficiency of our CoBayes. Using the same features, LASSO obtains the worst performance in terms of the three metrics since it only conducts linear regression using L1-regularization. Ridge using L2-regularization obtains a slightly better performance, which reflects the potential multicollinearity of features. We also notice that SVR and DT

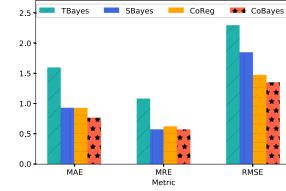


Fig. 10. Comparison between different components.

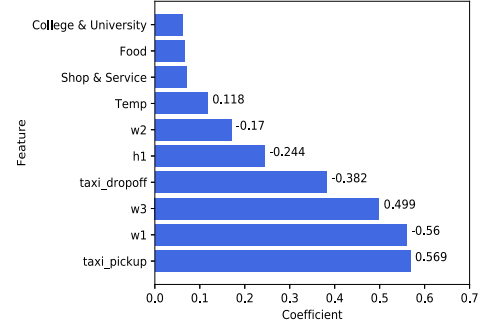


Fig. 11. Coefficients of top ten features.

achieve comparable performance with a slight difference. NB performs better than these two methods and LSTM in terms of RMSE and worse in terms of the other two metrics. This may be because the whole dataset does not fit an NB distribution well. LSTM obtains lower RMSE value with higher MAE and MRE values compared with DT. Note that RF obtains comparable performance with CoBayes in terms of MAE and MRE, verifying its strong learning ability. For RMSE, CoBayes achieves better performance than all the methods. These results may indicate the difficulty in a precise estimation with such a large dataset.

We also compare our CoBayes with TBayes, SBayes, and CoReg by using different components in Fig. 10. SBayes performs much better than TBayes, indicating the significance of spatial features in this problem. Compared with TBayes and SBayes, the CoReg performs better by exploiting the spatio-temporal features in a cotraining approach, which means the importance of both sets of features. CoBayes further improves performance by introducing near-repeat features. The improvement indicates the importance of recurrence risk. In general, our method can infer the crime risk in different roads and time durations successively.

To further interpret the importance of features in this article, we conduct a feature significance study using Pearson and Spearman correlation analysis on all features with continuous and discrete data, respectively, w.r.t crime risk values. We normalize values of all features to make them comparable. From Fig. 11, we observe that taxi mobility features, including *taxi_pickup* and *taxi_dropoff*, play important roles in affecting crime events. Within them, the feature *taxi_pickup* is the most important feature, representing the leaving mobility of that region. For example, when crime events happen on a road, this information would propagate to adjacent roads, leading to more people leaving the region in any means, certainly including taking taxis. In contrast, *taxi_drop* feature plays an important and negative effect. A probable reason is that consumers may take a taxi to home, office, and other necessary and urgent locations. These consumers have a certain degree in property, and the possible destinations may be well-protected neighborhoods, well-equipped offices, and other necessary and urgent locations. Besides, time duration *w1* is

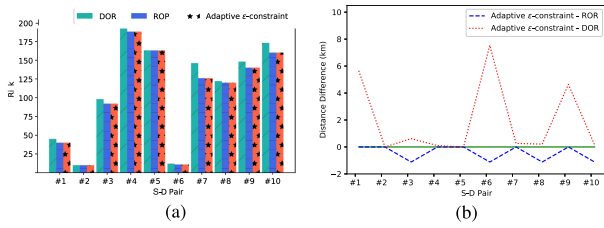


Fig. 12. Performance comparison of ten pairs of source and destination. (a) Risk comparison. (b) Distance comparison.

a comparably important feature, which indicates the temporal relationship with the occurrence of crime events. It should be noticed that both $w1$ and $h1$ play noticeable negative effects while $w3$ plays a strong positive effect. It indicates that crime events have a higher probability of happening at night, rather than wee hours. Besides time durations, air temperature also has a polynomial relationship with crime occurrences, as consistent with the demonstration in Fig. 4(b).

E. Evaluation on Risk-Aware Route Recommendation

We evaluate our method's performance by comparing it with two methods, only considering their singular objective function, i.e., ROR and DOR. Crime is a low-frequency event. There would be a few roads that may witness crime occurrences in a given time duration, while many roads would not witness any crime. Thus, in the experiments, we randomly select ten source and destination pairs from locations with frequent crime occurrences. Here, these locations refer to intersections of roads where at least one crime occurs during a day at average. From Fig. 12(a), we observe that DOR obtains higher crime risk compared with the other methods because it only considers route distance and sacrifices potential crime risk consideration of the roads. Our method achieves approximately equal risk with ROR since both methods choose to give priority to crime risk. For the distance comparison, we present two distance differences, which are equal to the distance computed by our method minus the distance computed by ROR or DOR, respectively. From Fig. 12(b), we can see that DOR obtains the lowest distance, while ROR obtains the highest distance, among most cases. Our method achieves comprehensive results since it considers crime risk and distance together. For example, for the pair #3, the route generated by our method has lower crime risk compared with DOR and lower distance compared with ROR. Note that for some cases, e.g., pair #2 and pair #5, routes generated by all three methods have equal crime risk and distance. It may be because roads around the source or destination location have potential crime risk, while the intermediate roads do not sense crime risk. Thus, the three methods may generate the same results on the route recommendation.

F. Runtime Performance

We implement the proposed CoBayes and adaptive ϵ -constraint methods using Python, which provides convenient tools and libraries for data preprocessing, analysis, model construction, and results visualization. We deploy our framework on a server with Intel Xeon E5-2609 CPU and 32-GB RAM, and it takes an average of 8.82 s to do one round of crime risk inference and 1.44 s to do one round of recommended routes generation, respectively.

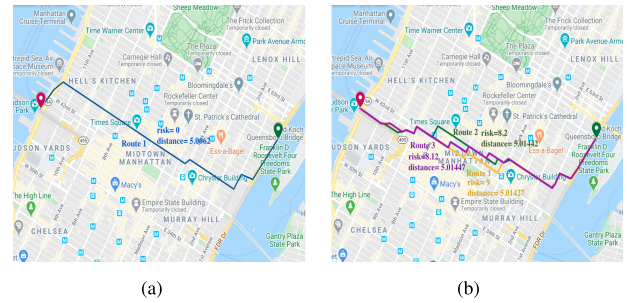


Fig. 13. Case study on route recommendation in Manhattan. (a) During 1:00–7:00 on weekdays. (b) During 12:00–21:00 on weekdays.

G. Case Study

We conduct a case study on crime risk-aware route recommendation in different time durations to present the different route suggestions according to different inferred crime risks. Fig. 13 shows the crime risk-aware route recommendation of given source and destination during different time durations in Manhattan. In the past decades, Manhattan has witnessed a lot of crime events. We randomly select two locations, as a source near Jacob K. Javits Convention Center and a destination near Ed Koch Queensboro Bridge. We provide risk-aware route suggestions in different durations. As shown in Fig. 13(a), during the wee hours on weekdays (1:00–7:00), the route with the shortest distance is recommended. In Fig. 13(b), we can see that there are several different routes recommended during the active hours on weekdays (12:00–21:00), without the route recommend in Fig. 13(a). These routes are crime risk-aware routes with longer distances. The reason behind this may be the homeless shelters in W 45th St, which helps provide temporary residence for homeless individuals. Some homeless individuals living in homeless shelters have many arrest records. The roads near W 45th St may have potential crime risk in active hours rather than the sleeping time. Note that our experiments were conducted based on the data in 2015. In 2017, the city Mayor announced a plan to close shelters, e.g., Aladdin Hotel, in W 45th St which brings many crime issues, such as violence and drug use.

VII. DISCUSSION

We would conduct a discussion from the following aspects.

For the performance in hold-out samples, we can accurately predict the crime risk of 99.52% samples. However, there are several limitations of this article. First, the *data sparsity* affects the training of our model. 1) Crime is a low-frequency event. 2) Some crimes were unreported [51]. In this case, there are some samples in which we would overestimate or underestimate the crime risk. Second, the generalization ability of this framework is limited since the factors affecting crime differ among different counties/cities. Although our model performs well using the data of New York City, it cannot be directly transferred to other cities.

For the prediction failure issue, there should be some measures to cope with the cases when the model fails to predict the crime risk accurately. The prediction failure can be divided into two types, i.e., overestimation and underestimation. When we overestimate the crime risk, citizens will make more cautious decisions with the higher predicted crime risk information. When we underestimate the crime risk, citizens need more information to avoid making risky decisions. To cope with this situation, one suggestion is to develop a real-time crime risk

warning system, which can provide and spread timely warning messages to people around and police officers through apps and broadcasting.

For the safety of recommended routes, we notice that a smaller value of predicted crime risk indicates a safer route. For example, when the overall predicted road-level crime risk of one route equals zero, we regard the route is safe. Citizens can go through this route without any worries. When the predicted crime risk is greater than zero, e.g., 1, we consider that a harassment or disorderly conduct event may occur in the route. In this case, citizens can adjust their decisions depending on their concerns about the crime risk. When the predicted crime risk of the route is even more significant, e.g. 3, we regard the route as dangerous since there may be a robbery or several assault events.

VIII. CONCLUSION AND FUTURE WORK

In this article, we studied the problem of crime risk inference at road level, using large-scale heterogeneous urban data. Specifically, we exploit the spatio-temporal features extracted from heterogeneous sensing data, e.g., mobility data, infrastructure data, meteorological data, demographic data, etc. Considering the near-repeat phenomenon, we extract recurrence crime features from historical crime records. Then we feed spatio-temporal features into a cotraining model to obtain spatio-temporal crime occurrence inferences. Incorporating it with recurrence crime features, we infer the final knowledge of crime risk. To present this problem's applicability, we enable a novel application as a crime risk-aware route recommendation. To evaluate the effectiveness of our methods, we use real-world datasets from New York City. Experimental results show that our methods can infer crime risk accurately, outperforming other methods. Experiments on crime risk-aware route recommendations also verify that the crime risk-aware route suggestions are generated effectively. We also conduct case studies to present the significance and advantages of our methods.

Despite the abundant research results we obtained, there are still several limitations. First, due to the data access difficulties, we only collected taxi trip data to represent the mobility features. Thus, one open future work is to collect more data to enrich the mobility features, e.g., check-in data, bus data, and ride-sharing car data. Second, we did not consider that different people have different tolerance for crime risk. One of the future works will be to provide personalized crime risk-aware route recommendations.

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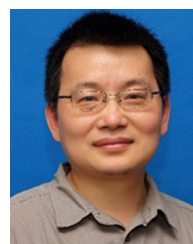
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